

Device Lock Service API

Filename	Device Lock Service API.docx	Reference	
Version	1.0.0	Date	
Status		Classification	Confidential

History

Status/Version/Date	Author	Comment
Stable/1.0.0/2025-10-21	Kenny. Zheng	First version.
Stable/1.0.1/2025-10-22	Kenny. Zheng	• Move algorithm and database to separate documents. • Add sync unlock API. • Update sync pin code API.
Stable/1.0.2/2025-10-22	Kenny. Zheng	Add parameters on sync clear/unlock API.
Stable/1.0.3/2025-10-28	Lei. Gao	Get Device Status API add errorCode.

1 Scope

This document specifies the device-side API interface design for the Device Lock Management Service.

2 Architecture

To avoid impacts on the DP Service, the Device Lock Management Service is deployed on the server side.

3 Specification

The Device Lock API complies with standard SaaS API specifications, including URL format, authentication mechanism and response format.

3.1 URL

The API URL complies with the following specifications:

```
https://<saas-host>/device-lock/api/<api-version>/<api-path>
```

SaaS Hosts are listed in the table below:

Environment	Host
DEV	dev-saas.zeasn.tv
ACC	acc-saas.zeasn.tv
PROD	saas.zeasn.tv

The API Version is currently fixed as: v1.

Example:

```
https://dev-saas.zeasn.tv/device-lock/api/v1/device/status
```

3.2 Authentication

Authentication is required for all requests to the Device Lock API. For the authentication algorithm, please refer to:

The developer obtains the access/secret-key from zeasn SAAS platform, if an interface require authorization, he needs to authorize the request in the following way:

```
ts = String(getCuurentTimestampInMilliseconds());
encryptStr = RequestURL + ts;           //RequestURL sample: /device-lock/api/v1/device/status
signature = Base64(HMAC_SHA1(SecretKey, encryptStr));
authorizationValue = AccessKey + ":" + signature + ":" + ts;
add a request header named "Authorization" with a value of authorizationValue
```

If the following conditions occur, the server will return http 401 error:

- A、 Authorization header not found
- B、 "ts" differs from the server timestamp by more than 1 hour
- C、 "signature" verification failed
- D、 The corresponding requestURL(API) is not authorized

3.3Response

The API response adopts the standard SaaS JSON response format. For details, please refer to:

The return value of all interfaces is in JSON format, and the format is as follows:

Param	Description
errorCode	Error code
errorMsg	Error information. When errorCode is not 0, this field has a value
data	Data content. The specific format depends on the corresponding interface
totalSize	The total number of records. When the interface supports paging, this field has a value
timestamp	Server timestamp, milliseconds

Common Error Code

The general error codes are as follows. For business-related error codes, please refer to the specific interface.

Error Code	Description
0	success
500	internal server error
1001	illegal argument error
1002	no authority

4 API List

4.1 Get Device Status

API Path	/device/status
Description	Retrieve device status, and return the specified error code if the device does not exist.

Parameters:

Name	Required	Comment
eui64	Y	Device EUI64

Result:

Field	Type	Description
paymentStatus	Integer	Payment Status: 0- In Progress 1- Completed
lockStatus	Integer	Lock Status: 0- Unlocked 1- Locked
clearStatus	Integer	Clear Status: 0- Normal 1- Clearing

Example:

```
{
  "timestamp": "1761040688237",
  "errorCode": "0",
  "errorMsg": null,
  "data": {
    "paymentStatus": 0,
    "lockStatus": 1,
    "clearStatus": 0
  }
}
```

ErrorCode:

ErrorCode	Description
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260001	Device does not exist
--------	-----------------------

4.2Sync Clear Status

API Path	/device/cleared
Description	Notify the server that the device clearing is completed. The server will reset device-related data.

Parameters:

Name	Required	Comment
eui64	Y	Device EUI64
clearTime	N	Timestamp of device clearing completion (in milliseconds). If empty, the server will use the current time.

Result: N/A

4.3Sync Unlock Status

API Path	/device/unlocked
Description	Notify the server that the device unlocking is completed. The server shall update the lock_status of the device.

Parameters:

Name	Required	Comment
eui64	Y	Device EUI64
unlockTime	N	Timestamp of device unlocking completion (in milliseconds). If empty, the server will use the current time.
unlockPassCode	N	PassCode used for unlocking.

Result: N/A

4.4Get Pin Code Group

API Path	/device/pinCode
Description	Retrieve available Pin Code groups for the device. The server only returns online Pin Code groups with the usage status of "unused".

Parameters:

Name	Required	Comment
------	----------	---------

eui64	Y	Device EUI64
-------	---	--------------

Result: Array<PinCodeGroup>

PinCodeGroup:

Field	Type	Description
codeId	Long	
passCode	String	
pinCode	String	

Example:

```
{
  "timestamp": "1761040688237",
  "errorCode": "0",
  "errorMsg": null,
  "data": [{
    "codeId": 1,
    "passCode": "T5ZX",
    "pinCode": "96SK4M"
  }, {
    "codeId": 2,
    "passCode": "5PZE",
    "pinCode": "B5R8PL"
  }, {
    "codeId": 3,
    "passCode": "HP6J",
    "pinCode": "WL6AR1"
  }]
}
```

4.5 Sync Pin Code Status

API Path	/device/pinCode/used
Description	Synchronize the used Pin Code groups.

Parameters:

Name	Required	Comment
eui64	Y	Device EUI64
passCodes	Y	List of used Pass Codes, separated by commas (,). Example: "ABCD,EFGH"

Result: N/A

5 ErrorCode List

Error Code and Description

ErrorCode	Description
260001	Device does not exist